

Introduction to Control Systems

Open-loop vs. closed-loop: the two ways to make a system behave.

Ogata, *Modern Control Engineering*, Ch. 1.

This is the starting point for the whole collection. In this tutorial you will:

- contrast **open-loop** and **closed-loop (feedback)** control,
- see why feedback rejects disturbances that open-loop control cannot, and
- meet the canonical unity-feedback block diagram used throughout.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A **control system** is an interconnection of components that regulates its own behavior or that of another system, typically to achieve a desired output. The two fundamental architectures are **open-loop** and **closed-loop (feedback)** control.

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Open-Loop Control

An open-loop system computes its actuating signal from the reference input alone, with no measurement of the actual output:

$$U(s) = G_c R(s)$$

$$Y(s) = G(s)G_c R(s)$$

Open-loop control is simple and stable (no feedback loop to destabilize), but it has **no mechanism to correct for disturbances or plant uncertainty** – the output tracks the reference only as well as the model $G(s)$ is known.

Closed-Loop (Feedback) Control

A closed-loop system measures the output $y(t)$, forms the error $e(t) = r(t) - y(t)$, and drives the actuator from that error:

$$\frac{Y(s)}{R(s)} = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)H(s)}$$

where $H(s)$ is the sensor/feedback transfer function (often $H = 1$ for unity feedback). Feedback trades some complexity and the possibility of instability for the ability to **automatically correct for disturbances and modeling error**, provided the loop is properly designed.

Example: Plant Subject to a Load Disturbance

Compare open-loop and closed-loop (unity-feedback, proportional control) responses to a step reference **and** a step disturbance entering at the plant input, for $G(s) = \frac{1}{s+1}$.

```
G = tf(1,[1 1]);
Kp = 9;    % proportional gain for the closed-loop case

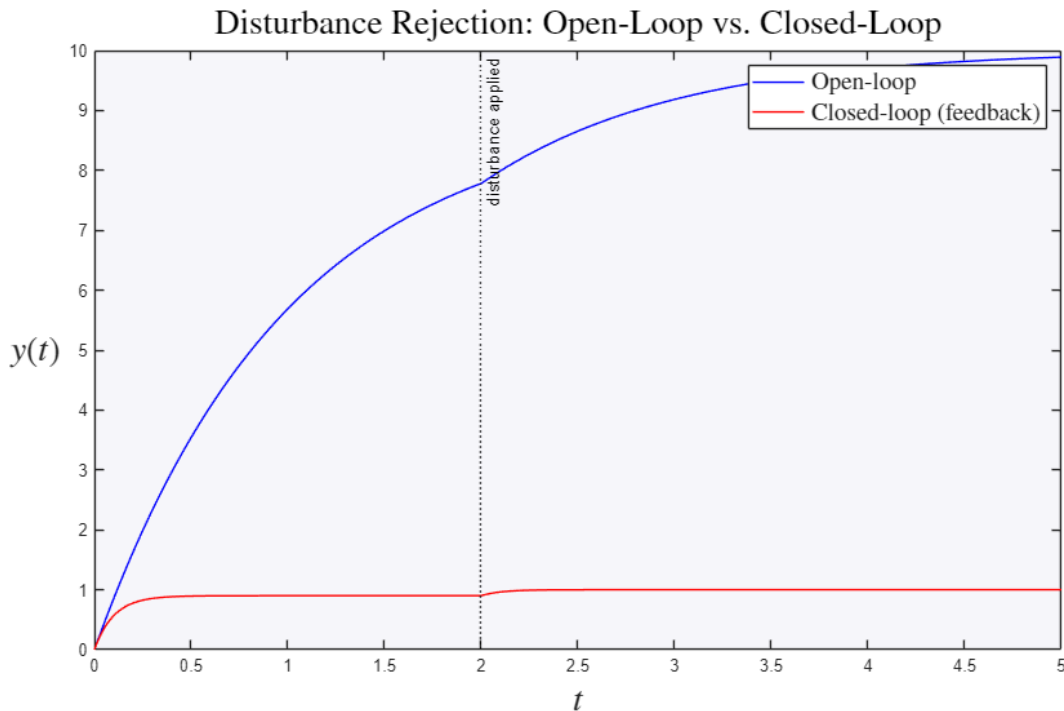
% Open-loop: u = Kp*r directly, no correction for disturbance d
sys_ol_r = Kp*G;    % reference path
sys_ol_d = G;    % disturbance path (unity gain into plant)

% Closed-loop: e = r - y, u = Kp*e + d enters at plant input
sys_cl_r = feedback(Kp*G,1);    % reference -> output
sys_cl_d = feedback(G,Kp);    % disturbance -> output

t = 0:0.01:5;
r = ones(size(t));
d_step_time = 2;    % disturbance step applied at t=2
d = double(t>=d_step_time);

y_ol = lsim(sys_ol_r,r,t) + lsim(sys_ol_d,d,t);
y_cl = lsim(sys_cl_r,r,t) + lsim(sys_cl_d,d,t);

figure
plot(t,y_ol,'b',t,y_cl,'r')
hold on
xline(d_step_time,'k:', 'disturbance applied')
hold off
legend('Open-loop','Closed-loop (feedback)','Interpreter','latex','FontSize',14)
title('Disturbance Rejection: Open-Loop vs. Closed-Loop','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



The open-loop response permanently shifts after the disturbance hits (nothing measures or corrects for it); the closed-loop response is pulled back toward the reference because the error signal $e = r - y$ picks up the deviation and the controller acts on it.

```
ess_ol = y_ol(end) - 1;
```

```

ess_cl = y_cl(end) - 1;
fprintf('Steady-state error after disturbance -- open-loop: %.4f, closed-loop: %.4f\n', ...
    ess_ol, ess_cl)

```

```

Steady-state error after disturbance -- open-loop: 8.8896, closed-loop: -0.0000

```

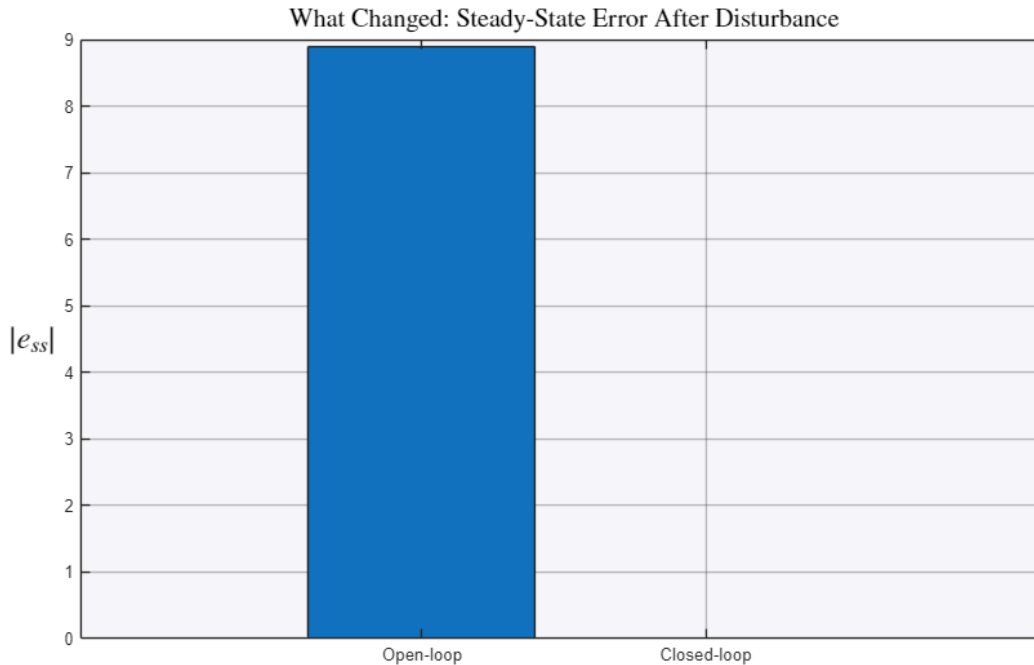
What Changed: Disturbance-Induced Steady-State Error

Quantify the "after the disturbance" error for each architecture: the open loop passes the full disturbance through to the output, while feedback rejects most of it – the whole reason to close the loop.

```

figure
bar([abs(ess_ol) abs(ess_cl)])
set(gca,'XTickLabel',{'Open-loop','Closed-loop'})
grid on
title('What Changed: Steady-State Error After Disturbance','Interpreter','latex','FontSize',15)
ylabel('$|e_{ss}|$', 'Interpreter','latex','FontSize',18)
set(get(gca, 'YLabel'), 'Rotation', 0)

```



Canonical Feedback Block Diagram

The standard unity-feedback diagram analyzed throughout this repository is: reference $R(s) \rightarrow$ summing junction (error $E = R - Y$) $\rightarrow$ controller $G_c(s) \rightarrow$ plant $G(s) \rightarrow$ output $Y(s)$, fed back to the summing junction. MATLAB's 'feedback' function implements exactly this reduction:

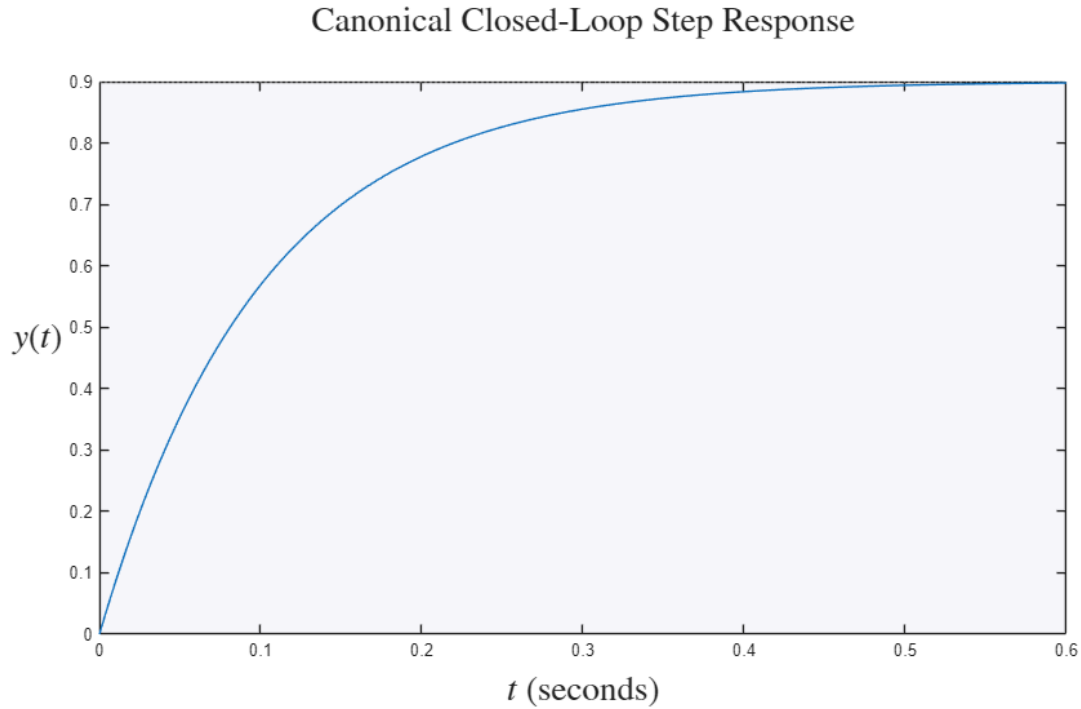
$$T(s) = \text{feedback}(G_c G, 1) = \frac{G_c G}{1 + G_c G}$$

```

Gc = tf(Kp,1);
T = feedback(Gc*G,1);
figure
step(T)
title('Canonical Closed-Loop Step Response','Interpreter','latex','FontSize',20)

```

```
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



This canonical loop – and its state-space, root-locus, and frequency-domain analysis counterparts – is developed in depth in the topic directories of this repository: [Mathematical Models/](#), [Transient and Steady-State/](#), [Root-Locus/](#), [Frequency-Response/](#), [PID Controllers/](#), and [State-Space/](#).

Try it yourself

- Change the proportional gain K_p from 9 to 2 and re-run: notice the closed-loop response gets slower and the residual disturbance error grows – feedback helps **less** with less loop gain.
- Move the disturbance earlier (`d_step_time = 1`) and watch the open-loop curve shift permanently while the closed loop recovers.